2.1 Laws of Composition

- Any rule combining pairs $a, b \in S$ to get another element in S.
  ↓
  addition, multiplication, matrix mult...

- $S \times S \to S$

- Using multiplicative notation, a law of composition is associative if the rule

  $(ab)c = a(bc)$ holds for all $a, b, c \in S$.

- A law is commutative if $ab = ba$

- An identity for a law of composition is an element e of S such that

  $ea = a$ and $ae = a$ $\forall a \in S$

- There can be at most 1 identity otherwise $e' = e'$ ↯
  The identity element will often by denoted by 1 if the loc is written multiplicatively. 0 if written additively

- Suppose that a loc on S, written multiplicatively, is associative and has an identity 1. An element a of S is invertible if $\exists b$ s.t.

  $ab = 1$ and $ba = 1$ b is the inverse of a

  $x: a^{-1}$

  $+: -a$

Properties of inverses:

- if an element a has both a left inverse l and a right inverse r, i.e., if $la = 1$ and $ar = 1$, then $l = r$, a is invertible, r is its inverse.
- If $a \in S$ is invertible, its inverse is unique
- If a and b are invertible then $(ab)^{-1} = b^{-1}a^{-1}$
- If only left/right inverse, it is not invertible

2.2 Groups and Subgroups

Group is a set G together with a law of composition that has the following properties

- associative
- contains identity element
- $\forall a \in G$, a has an inverse

- An albelian group is a group whose law of composition is commutative.
  - nonzero reals under multiplication
  - $\mathbb{R}$ under addition

- The order of group G is the number of elements it contains. $|G|$

  If order is finite, it is said to be a finite group. Otherwise, G is an infinite group.
  - Some infinite albelian groups:
    - $\mathbb{Z}^+$ the set of integers with addition as its law of composition
    - $\mathbb{R}^+$ same as $\mathbb{Z}^+$ but with reals
    - $\mathbb{R}^\times$ reals with multiplication as its law of composition (nonzero)

Let M be the set of maps from a set T to itself. A map $f: T \to T$ has an inverse function if and only if it is bijective, in which case we say f is a permutation of T. The permutations of T form a group, the law being composition of maps.

- The group of permutations of the set of indices $\{1, 2, \ldots, n\}$ is called the symmetric group and is denoted by S_n.

There are $n!$ permutations of a set of n elements, so the symmetric group S_n is a finite group of order $n!$.

for 2 elements, the permutations are the identity i and the transposition τ.

Multiplication table for the group contains the products $11, 1g, g1$, and gg. $gg = 1$

For S_3, we have $|S_3| = 6$

To describe it we pick 2 permutations (123) and (12)

label x & y for 2, 3.

(123)(12) 1 2 3

= (13)(2) 3 2 1

A subset H of group G is a subgroup if it has the following properties.

- closure - identity - inverses

1.5 Permutations

A permutation of S is a bijective map p of set S to itself: $p: S \to S$

The table

i	1	2	3	4	5
$p(i)$	3	5	4	1	2

exhibits a permutation of the set $\{1, \ldots, 5\}$

$p(1) = 3 \ldots$

It is bijective because every index appears once in the bottom row.

Cycle notation: $p(3) = 4$, $p(4) = 1$, $p(1) = 3$

(341)



2 cycles (transposition) $p(2) = 5$ $p(5) = 2$

$p = (341)(25)$

cycle notation is not unique and order doesn't matter

(341)
(134)
(413)

$(25)(341)$

2.3 Subgroups of the Additive Group of Integers

A subset S of a group G with law of composition written additively is a subgroup if it has these properties

- closure, if $a, b \in S$, then $a + b \in S$
- identity, $0 \in S$
- inverses, if $a \in S$ then $-a \in S$

Let a be an integer different from 0.

$$\mathbb{Z}a = \{ n \in \mathbb{Z} \mid n = ka \text{ for some } k \text{ in } \mathbb{Z} \}$$

integers divisible by a

This is a subgroup of $\mathbb{Z}^+$.

Thm 2.3.3
Let S be a subgroup of the additive group $\mathbb{Z}^+$. Either S is the trivial subgroup $\{0\}$, or else it has form $\mathbb{Z}a$, where a is the smallest positive integer in S.

The set of all integer combinations $ra+sb$ of a and b is
$S = \mathbb{Z}a + \mathbb{Z}b = \{n \in \mathbb{Z} \mid n = ra+sb \text{ for some integers } r, s\}$
is a subgroup of $\mathbb{Z}^+$. It is called the subgroup generated by a and b because it is the smallest subgroup that contains a and b.

2.3.3 tells us $\exists$ S which has the form $\mathbb{Z}d$, $d \in \mathbb{Z}$.
Generator d is the greatest common divisor of a and b, denoted as $gcd(a,b)$

Proposition 2.3.5: Let $a, b \in \mathbb{Z} - \{0\}$, and let $d = gcd(a,b)$. The positive integer that generates the subgroup $S = \mathbb{Z}a + \mathbb{Z}b$. So $\mathbb{Z}d = \mathbb{Z}a + \mathbb{Z}b$. Then
a) d divides a and b
b) If e divides a and b, it also divides d
c) There are integers r and s such that $d = ra+sb$

You can compute gcd by repeated division with remainder:
$a = 314, b = 136$
$314 = 2 \cdot 136 + 42$
$136 = 3 \cdot 42 + 10$ (Euclidean Algorithm)
$42 = 4 \cdot 10 + 2$
$10 = 5 \cdot 2$

$\mathbb{Z}(314) + \mathbb{Z}(136) = \mathbb{Z}(136) + \mathbb{Z}(42)$
$gcd(314, 136) = gcd(136, 42)$
$gcd(136, 42) = gcd(42, 10) = gcd(10, 2) = 2$

Two nonzero integers a and b are said to be relatively prime if the only positive integer that divides both of them is 1.

$\mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}$

Corollary 2.3.6: A pair $a, b \in \mathbb{Z}$ is relatively prime if and only if there are integers r and s such that $ra + sb = 1$.

(2,3) $(-1)2 + (1)3 = 1$

Corollary 2.3.7: Let p be a prime integer. If $p \mid ab$ then $p \mid a$ or $p \mid b$.

Proposition 2.3.8: Let $a, b \in \mathbb{Z} - \{0\}$, and let m be their least common multiple - the positive integer that generates the subgroup $S = \mathbb{Z}a \cap \mathbb{Z}b$. So $\mathbb{Z}m = \mathbb{Z}a \cap \mathbb{Z}b$. Then

a) $m \mid ab$

b) In $n \in \mathbb{Z}$, $n \mid a$ and $n \mid b$, then $n \mid m$.

Corollary 2.3.9: Let $d = \gcd(a,b)$ and $m = \text{lcm}(a,b)$. Then $ab = dm$

2.4 Cyclic Groups

The cyclic subgroup H generated by x is the set of all elements that are powers of x: (multiplication)

$H = \{\ldots x^{-2}, x^{-1}, 1, x, x^2, \ldots\}$

This is the smallest subgroup of G that contains x, denoted by $\langle x \rangle$.

Proposition 2.4.2 Let $\langle x \rangle$ be the cyclic subgroup of a group G generated by an element x, and let S denote the set of integers k s.t. $x^k = 1$.

a) The set S is a subgroup of the additive group $\mathbb{Z}^+$.

b) Two powers $x^r = x^s$, with $r \geq s$, are equal iff $x^{r-s} = 1$. $r - s \in S$.

c) Suppose that S is not the trivial subgroup. Then $S = \mathbb{Z}n$ for some positive integer n. The powers $1, x, x^2, \ldots, x^{n-1}$ are the distinct elements of the subgroup $\langle x \rangle$, and the order of $\langle x \rangle$ is n.

↳ the group $\langle x \rangle = \{1, x, \ldots, x^{n-1}\}$ described of this is called the cyclic group of order n.

The cyclic subgroup $\langle x \rangle$ generated by x has order n

Proposition 2.4.3: Let x be an element of finite order n in a group, and let k be an integer that is written as $k = nq + r$ where q and r are integers and r is in the range $0 \leq r < n$.

$x^k = x^r$

$x^k = 1$ iff $r = 0$

$d = \gcd(k, n)$. The order of x^k is equal to $\frac{n}{d}$.

What are Isometries?

└ You have a shape. You move the shape in some way. The shape remains rigid, the distance between 2 points on the shape remains the same.

- All isometries are either rotation, reflections, or translations. Isometries are unique
- M_n: the set of all isometries of $\mathbb{R}^n$
-